

Online Appendix – Not for Publication

A.1 Urbanization Data

We use two data series for urbanization rates. De Vries' (1984) figures are often considered the most reliable dataset for historical European urbanization. De Vries reports the share of population living in cities larger than 10,000 inhabitants. His data is available after 1500. Second, we derive urban population from Bairoch et al. (1988), who report the size of 2,191 European cities over the period 800-1850. As Bairoch et al. (1988) emphasize, estimates before 1300 are rough and less reliable. For many cities, observations are missing in at least some years. For example, Douai in France is reported to have 10,000 inhabitants in 1500 and 13,000 in 1700; no observation is given for 1600. We use linear interpolation to fill these gaps. When numbers are missing in 1400, we extrapolate the 1300 value with the country-specific population loss between 1300 and 1400, in order to approximate the impact of the Black Death in the mid-14th century.¹ For Cordoba (Spain) in 1000, we use population from Glick (1979), as proposed by Buringh and van Zanden (2009), correcting the unrealistically large number in Bairoch et al. (1988). These steps result in a comprehensive measure of urban population in Europe, which we divide by the population estimates of McEvedy and Jones (1978) to calculate urbanization rates at the country level, as used in figure 10.² Finally, in order to provide comparability with our first data series for European averages, we restrict the Bairoch et al. sample to countries covered by De Vries (which means excluding Russia and the Balkans). All urbanization rates that we report reflect weighted averages (by country population).

A.2 Wages and Urbanization

To derive the relationship between wages and urbanization, we pair income data from Maddison (2007) with urbanization rates from De Vries (1984). Figure A.1 plots urbanization rates in Europe and England in the early modern period against per capita income. The latter is normalized to unity for the pre-plague period.³ Our calibration of $\alpha = 0.68$ (see step 1 in section 4.1) traces out the pattern in the data.

[Insert Figure A.1 here]

A.3 Percentage of European Population Affected by Warfare

To derive the share of the European population that was affected by warfare, we use data on war-related deaths and epidemics from Levy (1983). His data show that, in a typical year, more than one European war was in progress – there were 443 war years during the period 1500-1800, normally involving three or more powers. Since it is the movement of armies, and not just military engagements that caused death, we count

¹The underlying country-level population data are from McEvedy and Jones (1978). For a number of countries where hard data is not available, McEvedy and Jones give estimates of population. We use the descriptive evidence in Benedictow (2004) to augment the data in McEvedy and Jones. For example, for the case of Norway, McEvedy and Jones do not give an estimate for population in 1400. Benedictow emphasizes the similarity of the plague's impact in Norway and England. Hence, we assume that the same percentage decline in population that occurred in England also affected Norway.

²Acemoglu et al. (2005) use the same approach to calculate European urbanization rates. However, they do not correct the raw data of Bairoch et al. (1988).

³Note that at this point $n_M = 2.5\%$ in the model, which follows from our calibration of initial conditions in step 2 in section 4.1.

the territories of combatant nations as affected if they were the locus of troop movements. We combine this information with demographic data in Maddison (2007). As European borders changed substantially over time, we use figures on population for the territory of states as they exist today. While anachronistic, this method has the advantage that it is not affected by border changes. We count all countries at war as affected because troop movements also occur within countries even when there is no fighting on their own soil. We sum the population in all affected countries for a given year, and divide by the corresponding total European population.⁴ Figure A.2 shows the resulting 10-year average together with a trendline.

[Insert Figure A.2 here]

A.4 Urbanization, Tax Revenues, and Warfare

What is the relationship between urbanization and state revenue in early modern Europe? And how does the latter affect the scale of warfare? These questions are crucial for the 'Horsemen effect.' In this section, we calibrate fundamental parameters for both relationships. To do so, we marry our urbanization data from section A.1 with the standard data on battles from Jaques (2007) and the information on government revenue from Karaman and Pamuk (2010). To reflect the scale of warfare, we use data on battles in early modern Europe. The number of battles can serve as an indicator of war intensity – a long war involves more battles than a short one. Counting wars alone would assign the same weight to the Thirty Years War and the War of Jenkin's Ear (1738-42) – clearly not a sensible choice.

We begin by estimating the relationship between tax revenues and urbanization implied by (25). Rearranging yields the non-linear estimation equation:

$$\frac{T_{it}}{\underline{c}N_{it}} = \tau n_M \frac{(1 - \alpha)}{1 - \alpha - n_M}, \quad (\text{A.1})$$

where the left-hand side variable is tax revenue per capita, normalized by the subsistence income $\underline{c}$. Tax revenues are measured in grams of silver, and the corresponding subsistence income is 205 grams.⁵ Note that (A.1) is only defined for $n_M < 1 - \alpha$, where $1 - \alpha = 0.32$ is the manufacturing share in above-subsistence consumption, as calibrated above. To avoid that extreme values close to the cutoff drive the estimation, we use only urbanization rates below 25%. Using non-linear least squares, we obtain the estimate $\hat{\tau} = 0.52$ with a standard error of 0.051 and an R^2 of 0.76. Figure A.3 shows the predicted relationship together with the data.

[Insert Figure A.3 here]

Next, we estimate the share of fixed costs in warfare. Equation (10) implies a linear relationship between the scale of warfare, W , and tax revenues, T , provided that the latter are large enough to cover the fixed

⁴Linear interpolation is used for the years where no population data are available.

⁵We calculate subsistence based on Maddison's figure of US-\$400. According to Maddison, the UK had a per capita income of \$714 in 1500. Since we know that the per capita tax (in silver) in 1509 was 5.5 grams (equivalent to a tax rate of 1.5% of average income), this implies that the average income was equivalent to 367 grams of silver per year - and US-\$400 correspond to 205 grams of silver per head and year.

cost F_W . The dataset represents observations in 25-year-intervals for Europe between 1500 and 1775. We estimate the equation $W_t = \beta_0 + \beta_1 T_t + \varepsilon_t$, where $\hat{\beta}_0$ can be interpreted as fixed cost of warfare. We make no claims about causation – our paper argues that it went both ways. F_W reflects the minimum cost of maintaining an organized body of soldiers, maintaining fortifications, keeping armories and supply depots ready, etc. The slope parameter is 19.99 (t-stat 14.39) with an intercept of $\hat{\beta}_0 = 197.95$ (t-stat 2.60) and an R^2 of .92. In the right panel of figure A.3, we show the fixed costs divided by total tax revenues ($\hat{\beta}_0/T_t$) for the various years. This fraction falls from about 80% in 1550 to approximately 20% in the 17th century.

In our model, the war- and trade-driven Horsemen are at work when tax revenues rise above the fixed cost F_W . We showed in section 4 (step 4) that this is the case for urbanization rates above 2%. An alternative way to write our model would be a threshold level $\underline{n}_M$ below which income is too low to be taxed or expropriated, serving merely to provide elementary nutrition and basic manufacturing goods. This would yield the same quantitative predictions as the fixed cost of warfare.⁶ We can check whether a calibrated version of this alternative model would involve similar magnitudes as our present approach. To calibrate $\underline{n}_M$, we use the data shown in figure 3. The corresponding regression line has an intercept of -1.71 (t-stat -1.93), and a slope of 55.84 (t-stat 9.98). These values imply that (per-capita) tax revenues exceed zero at $\underline{n}_M = 3.06\%$. Therefore, the fixed cost (normalized by population) must equal per-capita tax revenues at this point: $f_W = t(\underline{n}_M)$. Using (25), this implies: $f_W = \tau \underline{n}_M (1 - \alpha) \underline{c} / (1 - \alpha - \underline{n}_M) = 0.017$. This value is very similar to the 0.011 that we obtained in our baseline calibration, and it yields almost identical results.

A.5 Congestion and Constrained City Growth

Income increases raise the demand for urban goods and thus manufacturing wages, attracting migration to cities. However, in the short-run migration is constrained because new dwellings and infrastructure must be provided. Too many migrants therefore lead to over-crowding, making further migration to urban centers unattractive. In the interest of simplicity, we capture congestion effects with an upper limit to the growth rate of cities, $\bar{\nu}$. When shocks are large, and urban-rural wage differentials are substantial, this constraint becomes binding. It then takes time until population shares reach their steady state levels n_M^{LR} and n_A^{LR} , as given by the equation for unconstrained migration, (23).

Let N_A^* and N_M^* be the number of individuals living in the countryside and cities, respectively, at the beginning of a period. $N_M^{LR} = n_M^{LR} N$ denotes steady state urban population, i.e., the number of city inhabitants that would be established under unconstrained city growth if overall population is N . Next, we derive the growth of city population that occurs when migration is unconstrained, reaching the steady state

⁶We prefer the latter because of technical convenience. The threshold for taxation would introduce a discontinuity, with disposable income growing more slowly beyond $\underline{n}_M$. Our present way of modeling avoids this problem because it has a proportional tax on all manufacturing products.

instantly.⁷

$$\nu \equiv \frac{N_M^{LR} - N_M^*}{N_M^*} = \frac{n_M^{LR} - n_M^*}{n_M^*} \quad (\text{A.2})$$

The likelihood that congestion constrains migration is the larger the more the steady state population distribution deviates from actual values. If ν exceeds the upper bound for the growth rate of urban centers, the constraint $\bar{\nu}$ becomes binding. In this case, replacing ν with $\bar{\nu}$ in (A.2) gives the law of motion for city population:

$$N_M = (1 + \bar{\nu})N_M^* \quad (\text{A.3})$$

The remainder of the population works in agriculture: $N_A = N - N_M$. Wages, production, and relative prices under constrained city growth can be derived from the known location-specific employment. Note that urban and rural wages differ in this case. Agricultural wages are given by (4). The manufacturing wage depends on the relative demand for urban goods. Introducing location-specific wages in (2) together with market clearing yields an explicit solution for urban wages:⁸

$$w_M = \frac{1}{n_M} \frac{1 - \alpha}{\alpha} [w_A n_A - \underline{c}] \quad (\text{A.4})$$

Manufacturing products are sold at $p_M = w_M / [(1 - \tau)A_M]$. Workplace-specific food consumption follows from the corresponding wages and (2). Accordingly, baseline mortality as well as fertility also vary by location, following from (8) and (17), respectively. Aggregate birth and death rates are weighted averages of the rural and urban rates. The 'Horsemen effect' is calculated as in the unrestricted case. While none of the long-run (or qualitative) results in this paper depend on the assumption of congestion and limited city growth, it is important for historical realism in the transitional dynamics.

We assume that city growth is not constrained immediately after the Black Death, when the urban housing stock is partially empty. In particular, the constraint only sets in once *total* city population reaches its pre-plague level. The left panel of figure A.4 suggests that this was the case as early as in 1400. Note however, that overall population had not recovered yet by then – as the right panel of the figure shows, it was still well below its pre-plague level in almost all European countries. The combination of quickly rising numbers of urban dwellers and slower recovery of overall population implied a spike in urbanization rates in 1400 (as shown in figure 8). Once the pre-plague urban population was reached, new housing had to be built for further growth. That is when the migration constraint begins to bind in our model.

[Insert Figure A.4 here]

⁷More precisely, this is the growth rate of city population due to migration only. We implicitly assume that urban offspring do not contribute to congestion because they live with their parents, at least in the short run. In this specification, the growth rate of ν is equal to the growth of the urbanization rate – a fact that we use to calibrate $\bar{\nu}$.

⁸A detailed solution of the model with location-specific wages and demand is provided in the working paper Voigtländer and Voth (2008).

A.6 Steady State and Black Death in China

The left panel of figure A.5 shows the birth and death schedules for China. Note that the death schedule is slightly upward sloping only for high urbanization rates, which is due to the low belligerence parameter ($A_W^{\text{China}} = 0.25 \cdot A_W^{\text{Europe}}$). The economy has the unique steady state C , where fertility and mortality are 3%, and the urbanization rate is 2.2%.⁹ The right panel of figure A.5 shows the transitory effect of the Black Death in China, as described in section 4.5.

[Insert Figure A.5 here]

A.7 Robustness of Results for Europe and China

In section 4.6, we reported results for a lower-bound war-related 'Horsemen effect' of 0.5%. This corresponds to 1/6 of the baseline mortality at subsistence consumption. To derive the corresponding steady states, we leave all parameters unchanged, but cap $h_W(n_M)$ in equation (16) at 1/6. The left panel of figure A.6 shows the result. While the stable steady state E_0 and the unstable one E_U are unchanged, the high-income steady state E'_H now features lower urbanization rates – about 6 percent. Next, we show the results when we assume an upper bound for Chinese belligerence after 1400. The main calibration had $A_W^{\text{China}} = 0.25$. Here, we use $A_W^{\text{China}} = 0.5$ instead. The right panel of figure A.6 shows that even under this extreme assumption, China only exhibits the low-income steady state C . The reason is that with belligerence still substantially below the European level, the death rate rises too slowly to intersect with the birth rate from below.

[Insert Figure A.6 here]

This analysis provides important insights into the robustness of our mechanism: The maximum size of the 'Horsemen effect' is not essential for obtaining multiple steady states. On the other hand, the slope is crucial, i.e., the rate at which disease spread via warfare responds to income increases (and thus to urbanization). We argue that the high responsiveness in Europe reflects initial political conditions, where belligerent states used tax revenues almost entirely for warfare (Bean, 1973).

A.8 Negative Impact of Warfare on Productivity

As discussed in section 2, early modern warfare was destructive, but on a limited scale. Productivity suffered in the short-run, as a result of the destruction of physical capital, slaughter of livestock, and the disruption of communications. These adverse effects normally disappeared quickly once armies moved on.¹⁰ We now incorporate this negative impact into our model. In the modified setup, each country will be at war for four consecutive years in each decade. This is equivalent to the highest average observed for a fifty-year

⁹Note that the urbanization rate in steady state C is below the pre-plague level of 2.5% for Europe. The reason is that the direct 'Horsemen effect' – excess city mortality – is not at work in China, so that average mortality is slightly lower than in Europe, implying lower steady state urbanization. This difference becomes apparent when comparing C and E_0 in figure 6 in the paper.

¹⁰Even if warfare had a negative long-run impact on productivity, our main result would not change. Following the Malthusian logic, with permanently lower productivity levels, population would be lower in equilibrium. But p.c. incomes would still rise with mortality.

period in early modern Europe (see figure A.2). Under warfare, TFP decreases by 1% (5%), and returns to the baseline level when hostilities cease. Death rates increase by Δd . To isolate effects, we assume that neither city mortality nor trade add to background mortality in this exercise. The remainder of the decade is peaceful, and death rates are at their baseline level. During wars, we use conservative values, $\Delta d = 1\%$ and 2% , and simulate the model without migration constraints and without technological progress.¹¹ Figure A.7 shows the results.

[Insert Figure A.7 here]

Initially, the economy is in equilibrium C (see e.g. figure 6). Periodic warfare sets in after 50 years. The left panel shows the effect of negative TFP shocks of 1% during wars; the right one, 5%. In both cases, p.c. income rises and population falls in the long-run. Per capita income fluctuates more strongly when warfare is assumed to have a large negative effect on TFP, but overall development patterns are unaffected. With $\Delta d = 1\%$, p.c. income fluctuates around 1.21, which corresponds to an urbanization rate of 5.5%. For the case of larger mortality rates due to warfare ($\Delta d = 2\%$), it fluctuates around 1.34, which implies urbanization rates of 8.2% – the same as in our baseline calibration.

This modified analysis shows that our results are robust to relaxing two implicit assumptions. First, even with lower productivity under warfare, long-term p.c. income and urbanization rise. This is explained by the quick return of TFP to its pre-war level in early modern times, while population growth takes time to replace the deceased. Second, periodic warfare (with a large mortality increase during wars and zero additional mortality otherwise) delivers similar results as continuous warfare (with a constant average impact on mortality).

A.9 The European Marriage Pattern

Europeans curtailed fertility in an important way – by delaying marriage for most women, and ensuring that a high proportion never married. This pattern is known as the European Marriage Pattern (EMP), and it was particularly pronounced in North-Western Europe (Kusssmaul, 1981; Voigtländer and Voth, 2009, 2011). Lower birth rates for a given income have a similar effect as higher background mortality: Both alleviate population pressure and reduce the land-labor ratio in steady state. If EMP emerged as a consequence of the plague, as some authors argue (de Moor and van Zanden, 2010; Voigtländer and Voth, 2011), then some of the increase in North-Western European incomes after 1350 has to be attributed to the plague’s impact on fertility.

In the following, we investigate EMP’s contribution to city growth in our model. We focus on England, where births were unusually responsive to economic conditions (Lee, 1981; Wrigley and Schofield, 1981) in early modern times.¹² We also make the extreme assumption that birth rates were not responsive to income

¹¹These figures compare to the previous calibration as follows: With a maximum 38% of the European population affected by wars, 1 and 2 percent excess mortality during warfare translate into a 0.38 and 0.76 percent maximum war-related ‘Horsemen effect,’ respectively. The second value is therefore similar in magnitude to our baseline calibration.

¹²England was also ahead of the European average in terms of incomes and city growth: P.c. income grew by 75% between 1500 and 1700, and urbanization rates more than quadrupled from 3% to over 13% (and 20% in 1800) (Maddison, 2007; de Vries, 1984).

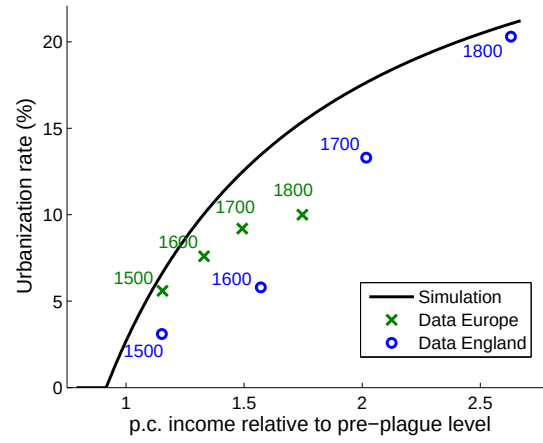
before the Black Death, so that $\varphi_b^{\text{before}} = 0$. This stacks the odds in favor of finding a larger impact of EMP, as to ensure that we do not exaggerate the role of the 'Horsemen effect.' The steady state before the plague is given by point S_0 in figure A.8. When EMP emerges after the plague, it shifts the birth schedule downwards by 1%. This corresponds to a 30% drop relative to the pre-plague steady state for a given p.c. income (or correspondingly, urbanization) level (Clark, 2007). EMP also makes birth rates responsive to income such that $\varphi_b^{\text{after}} = 1.4$, as in the baseline calibration.¹³

[Insert Figure A.8 here]

In the absence of the 'Horsemen effect,' the shift and tilt of the birth schedule move the economy from the pre-plague steady state S_0 to the S_{EMP} steady state; urbanization grows from about 2% to 7%. The 'Horsemen effect' creates an additional rise in urbanization (steady state S_{H+EMP}) – from 7 to 11 percent. Both effects appear to be equally important, and together they explain a large part of the observed increase in England's urbanization rate. This underlines the importance of the 'Horsemen effect' for increasing wages in early modern Europe – in continental Europe, where EMP was weaker, the Horsemen contribution was possibly even more significant.

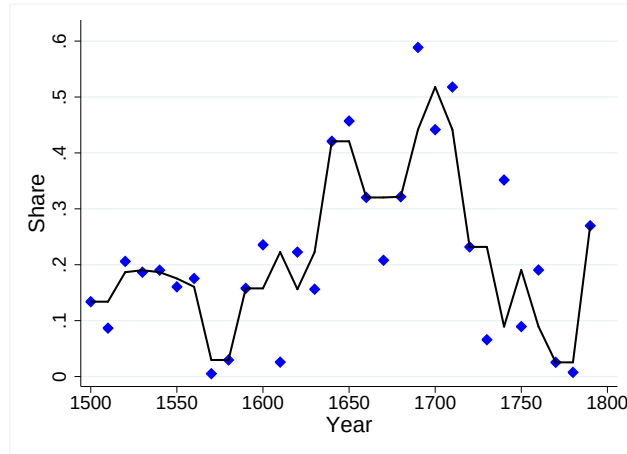
¹³The corresponding parameter values are $b_0^{\text{before}} = 0.03$ and $b_0^{\text{after}} = 0.02$. With all other parameters unchanged, this implies again a steady state urbanization rate of 2.5% before the plague.

Figure A.1: Urbanization and p.c. income – model vs. data



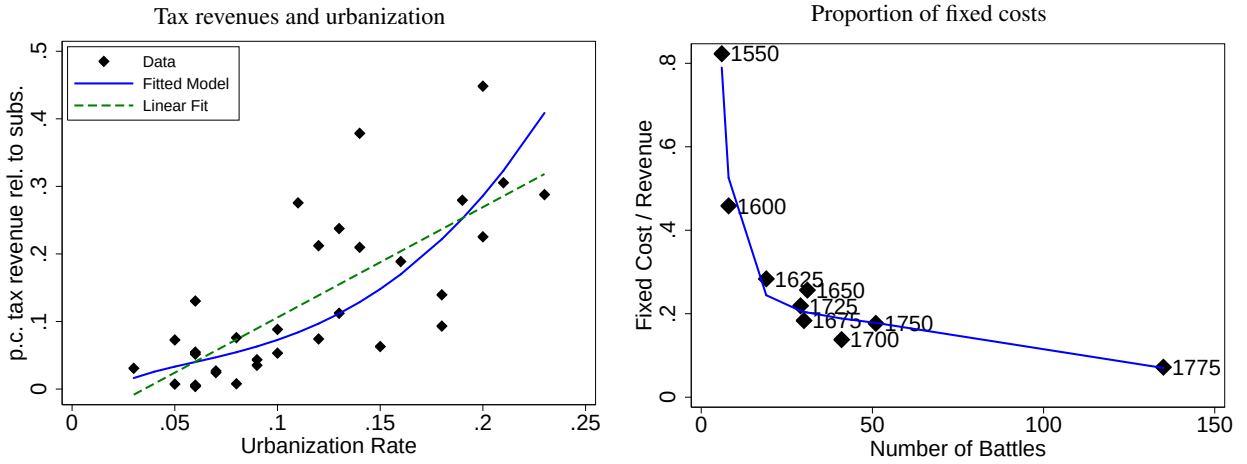
Data sources: Per capita income from Maddison (2007) is relative to the 1350 level (obtained with linear interpolation between 1000 and 1500). Urbanization rates from De Vries (1984).

Figure A.2: Percentage of European population affected by war



Data sources: War data from Levy (1983); population from Maddison (2007). Data points represent 10-year averages. The solid line is the median of each data point and the two adjacent ones.

Figure A.3: Calibration: Urbanization, tax revenues, and warfare

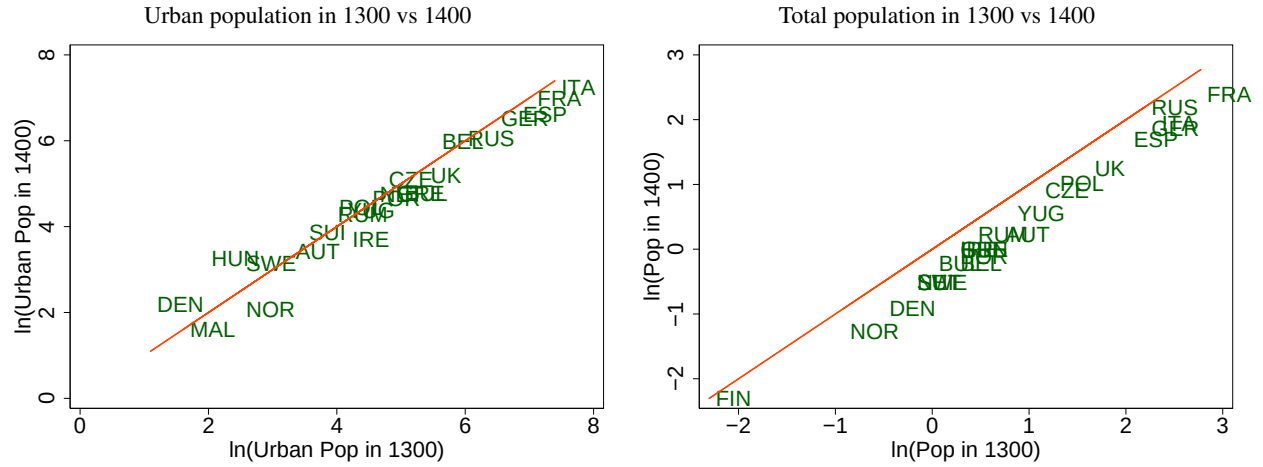


Notes: Urbanization rates: [Bairoch et al. \(1988\)](#) for population in cities larger than 5,000 inhabitants, divided by country-level population from [McEvedy and Jones \(1978\)](#); see appendix A.1 for details. Tax revenues (per capita in grams of silver) from [Karaman and Pamuk \(2010\)](#). P.c. tax revenue is normalized by subsistence income, which corresponds to 205 grams of silver (see footnote 5 for a detailed description). Data on the number of battles fought, by century and geographical area, is collected in [Jaques \(2007\)](#).

Left panel: The dataset contains 7 countries with observations in 50-year intervals between 1509 and 1759, as well as 1789. Because the estimation equation (A.1) restricts the range of urbanization rates, we only use data with below 25% urban population. For the estimated non-linear model, the regression line has a slope parameter of .52 with a t-statistic of 10.31 and an R^2 of .76. In the linear case the slope is 1.63 (t-stat 6.75) with an intercept of -.057 (t-stat -2.48) and an R^2 of .57.

Right panel: The fixed cost of warfare is obtained by running a regression of total tax revenues of the combatants in 374 war engagements in Europe between 1500 and 1775, in 25-year intervals. The slope is 19.99 (t-stat 14.39) with an intercept of 197.95 (t-stat 2.60) and an R^2 of .92. The estimate of the intercept represents the fixed cost of warfare. The vertical axis of the figure shows this intercept divided by total tax revenues for the various years.

Figure A.4: Urban and total population: 1300 vs 1400



Notes: Urban population from Bairoch et al. (1988) for cities larger than 5,000 inhabitants. Country-level population from McEvedy and Jones (1978); see appendix A.1 for details. Each figure plots a 45-degree line to compare outcomes in 1300 and 1400.

Figure A.5: Long-run impact of the plague with weak 'Horsemen effect' – The case of China

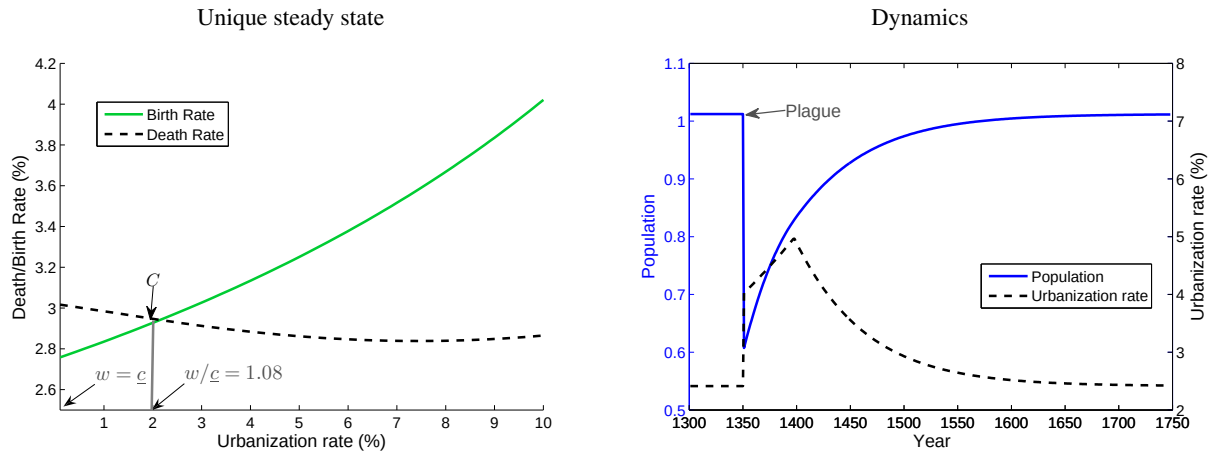


Figure A.6: Robustness of Results: Size of Horsemen effect and Belligerence

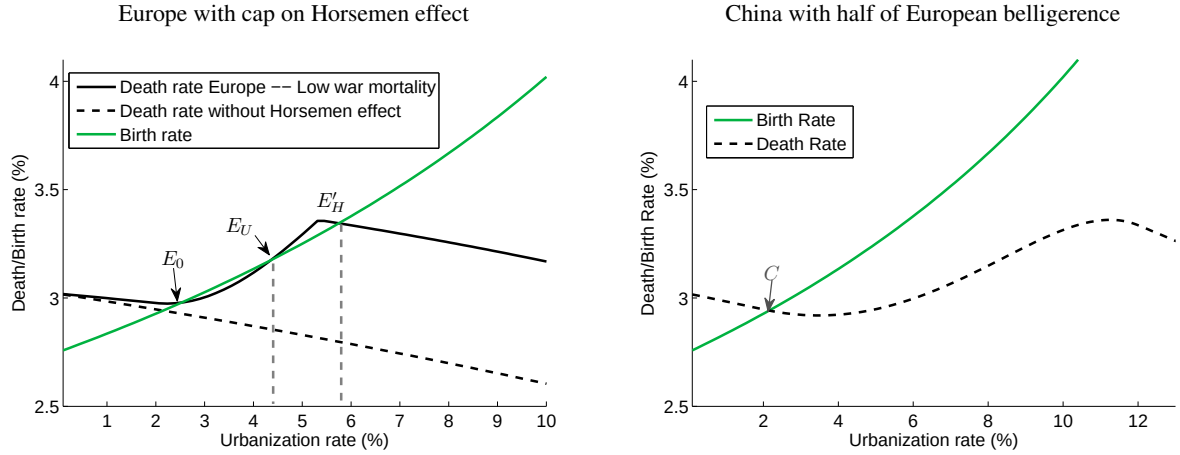


Figure A.7: Robustness of Results: Negative impact of warfare on TFP

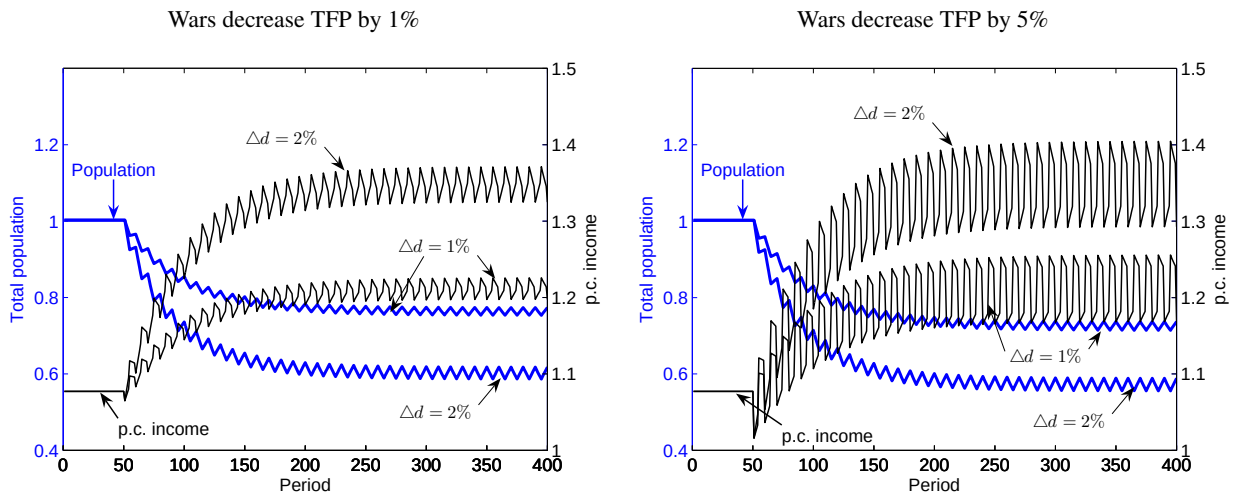


Figure A.8: Pre-and post-plague steady states with EMP and Horsemen

